

DE Lab7  
Hayden Fuller

Q1:

a) 31.5ps, 22.5ps, 18.75ps, 16.25ps

$$(30k/2)ohm*(2*3+3)*.1fF + (30k/2)ohm*(3*2)*.1fF + (30k/2)ohm*(3*2)*.1fF \\ = 13.5ps + 9ps + 9ps = 31.5ps$$

$$(30k/2)ohm*(2*3+3)*.1fF + (30k/2)ohm*(3*2)*.1fF \\ = 13.5ps + 9ps = 22.5ps$$

$$(3*12.5k/3)ohm*(2*3+3)*.1fF + (2*12.5k/3)ohm*(3*2)*.1fF + (12.5k/3)ohm*(3*2)*.1fF \\ = 11.25ps + 5ps + 2.5ps = 18.75ps$$

$$(3*12.5k/3)ohm*(2*3+3)*.1fF + (2*12.5k/3)ohm*(3*2)*.1fF \\ = 11.25ps + 5ps = 16.25ps$$

i) 40.5ps, 31.5ps, 26.25ps, 23.75ps

adds capacitance to the output of each, time dependant on resistance to that node

$$(2+1)*.2fF = 0.6fF$$

$$(30k/2)ohm*.6fF = 9ps \\ 31.5ps+9ps = 40.5ps$$

$$(30k/2)ohm*.6fF = 9ps \\ 22.5ps+9ps = 31.5ps$$

$$(3*12.5k/3)ohm*.6fF = 7.5ps \\ 18.75ps+7.5ps = 26.25ps$$

$$(3*12.5k/3)ohm*.6fF = 7.5ps \\ 16.25ps+7.5ps = 23.75ps$$

ii) 340.5ps, 331.5ps, 276.25ps, 273.75ps

same as part a.i,

$$100*.2fF = 20fF$$

$$15k*20fF = 300ps$$

$$12.5k*20fF = 250ps$$

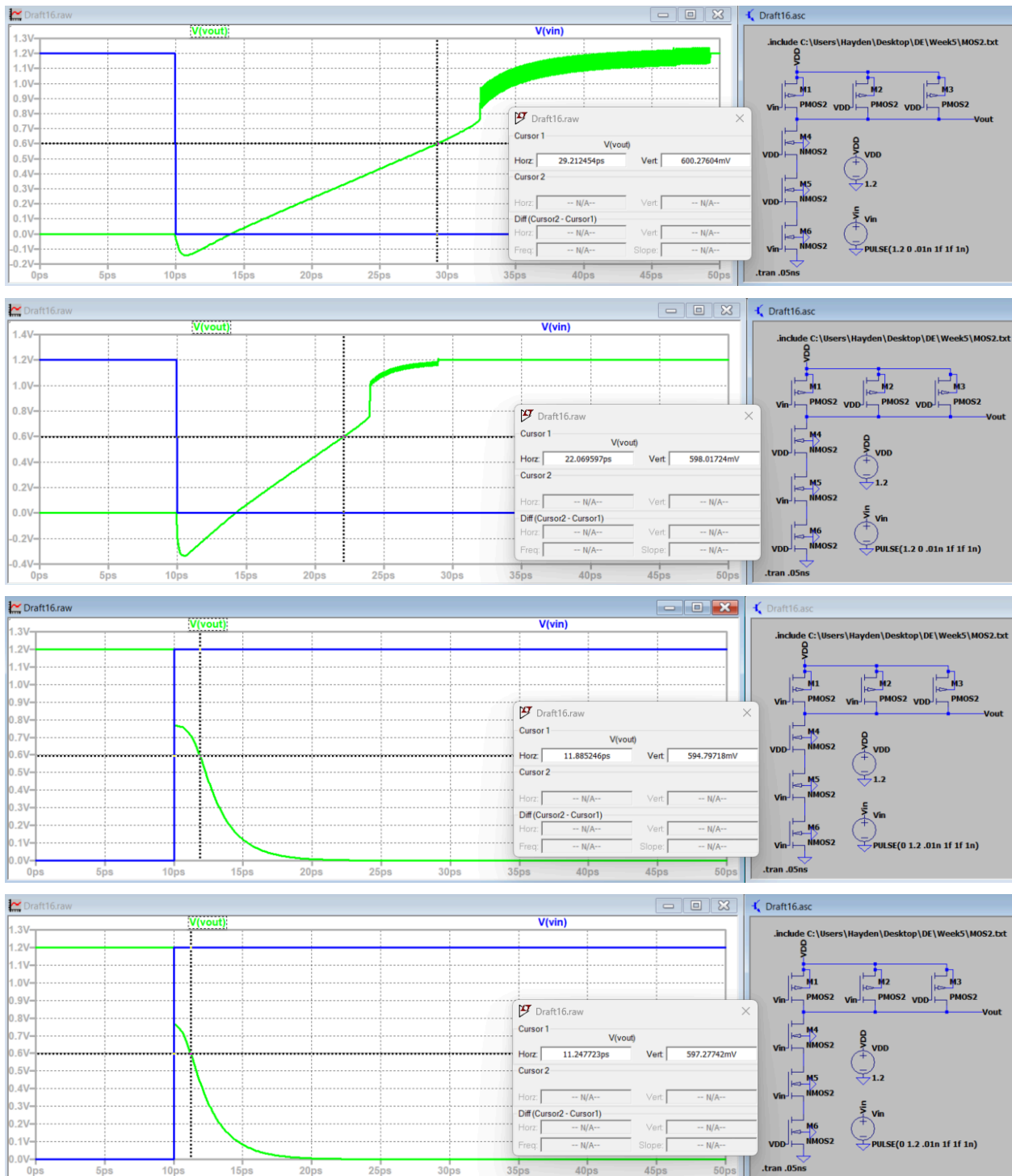
$$40.5+300=340.5ps$$

$$31.5+300=331.5ps$$

$$26.25+250=276.25ps$$

$$23.75+250=273.75ps$$

b) 29.21ps, 22.07ps, 11.89ps, 11.25ps



hand: 31.50ps, 22.05ps, 18.75ps, 16.25ps

LTspice: 29.21ps, 22.07ps, 11.89ps, 11.25ps

all values are a bit lower than my hand calculations, but overall quite close and following the same trends that falling is faster and which of the two options of each is faster

2)

a)

i) 0.75V

$$\Delta V_{\text{victim}} = \Delta V_{\text{aggressor}} (C_c / (C_{\text{gnd}} + C_c))$$

$$1.2 * (100\text{f}/60\text{f} + 100\text{f}) = 1.2/1.6 = .75\text{V}$$

ii) -0.75V

$$-1.2 * (100\text{f}/60\text{f} + 100\text{f}) = -.75\text{V}$$

b) 260fF

$$C_L = C_{\text{gnd}} + 2 * C_c$$

$$C_L = 60\text{f} + 2 * 100\text{f} = 260\text{fF}$$

i) 333%

$$t_{\text{opp}}/t_{\text{same}} = C_{\text{opp}}/C_{\text{same}} = 260\text{f}/60\text{f} = 4.33$$